Dilation

Ques ~~It is a~~

Binary Image

0	0	0	0	0
0	1	1	0	0
0	0	1	1	0
0	1	1	1	0
0	0	0	0	0

Structured image

1
1
1

Sol

Step 1:

→ Pad the binary image at top and bottom.

→ And then place overy structured image along with binary image and check for the condition.

↓

Fully match "1"

One match "1"

No match "0"

Note*:

Condition should be applied in structured element.

Step 2

top

0	0	0	0	0
0	0	0	0	0
0	1	1	0	0
0	0	1	1	0
0	1	1	1	0
0	0	0	0	0
0	0	0	0	0

bottom.

Structure image

1
1
1

Step : 3 final solution

0	0	1	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Erosion

Binary image

0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	0
0	0	0	0	0

Structured image

1
1
0

Step : 1

Condition

↳ Structured element should match completely with binary image, write "1". Otherwise "0".

Step : 2 final image

0	0	0	0	0
0	0	0	1	0
0	1	0	0	0
0	0	0	0	0
0	0	0	0	0

Opening

Que

A

0	0	0	0	0
0	1	1	1	0
0	0	0	1	0
0	1	0	1	0
0	0	0	0	0

B

1
1
1

Step : 1

formula $(A \ominus B) \oplus B$

erosion

dilation

Step : 2 we do erosion

0	0	0	0	0
0	0	0	1	0
0	0	0	0	0
0	0	0	0	0

note*:

Same Structured elem

Step : 3 to do dilation

→ add 0's on top & bottom

0	0	0	0	0
0	0	0	0	0
0	0	0	1	0
0	0	0	0	0
0	0	0	0	0

1
1
1

Step: 4 final ans:

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Q. Closing

A

0	0	0	0	0
0	1	1	1	0
0	0	0	1	0
0	1	0	1	0
0	0	0	0	0

B

1
1
1

Step: 1 formula

$$(A \oplus B) \ominus B$$

$\oplus$ → dilation

$\ominus$ → erosion

Step: 2

0	0	0	0	0
0	0	0	0	0
0	1	1	1	0
0	0	0	1	0
0	1	0	1	0
0	0	0	0	0
0	0	0	0	0

1
1
1

$A \oplus B$

Step: 3

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step: 4 now final erosion

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Step: 5 final answer

$$(A \oplus B) \ominus B$$

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$